

… fabriquer mes logarithmes comme on fabrique des épingles.

… to manufacture my logarithms as one manufactures pins.

Gaspard de Prony, *Notice sur les grandes tables logarithmiques…*, 1824. English: lecturer

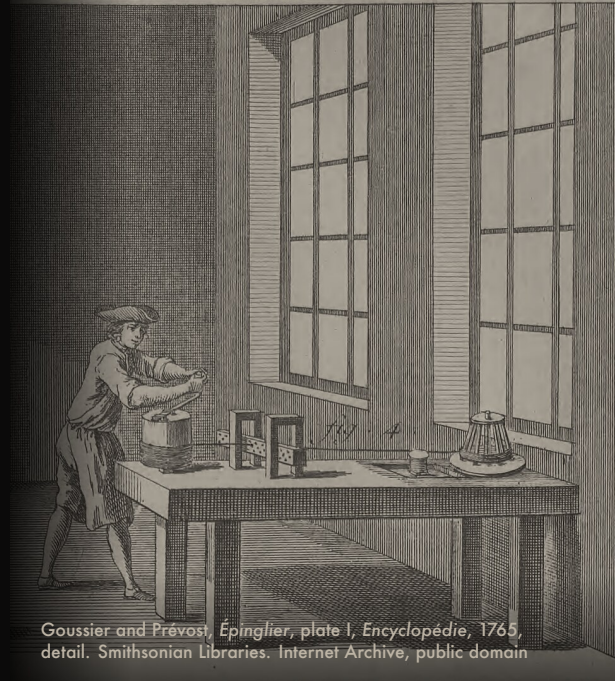
Goussier and Defehrt, *Épinglier*, plate II, *Encyclopédie*, 1765, detail. Smithsonian Libraries. Internet Archive, public domain

Adam Smith

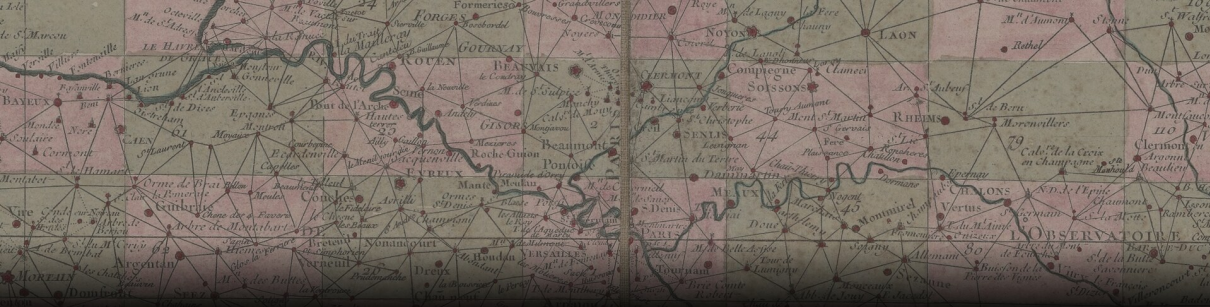
1723 to 1790

"One man draws out the wire; another straights it; a third cuts it; a fourth points it; a fifth grinds it at the top for receiving the head… Those ten persons, therefore, could make among them upwards of forty-eight thousand pins in a day."

Adam Smith, *The Wealth of Nations*, 1776, Book I, chapter 1



Goussier and Prévost, *Épinglier*, plate I, *Encyclopédie*, 1765, detail. Smithsonian Libraries. Internet Archive, public domain



Every parcel in France

The French Revolution, 1790 to 1793

The Revolution taxes land, so the area of every parcel has to be computed. Land is measured in triangles, computed with tables of logarithms.

In 1793 Gaspard de Prony, director of the Bureau du cadastre in Paris, is ordered to compute new tables.

Maraldi and Cassini de Thury, map of the principal triangles of France, 1744, detail. BnF. Wikimedia Commons, public domain

Three tiers

Bureau du cadastre, 1793 to 1796

Five or six mathematicians choose the formulas and pass them down.



The computers of the Harvard College Observatory, about 1890.
Wikimedia Commons, public domain

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Seven or eight analysts turn them into the first values of each page and pass the pages down.



The computers of the Harvard College Observatory, about 1890.
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Three tiers

Bureau du cadastre, 1793 to 1796

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Seven or eight analysts turn them into the first values of each page and pass the pages down.

About twenty computers fill in each page, using only additions and subtractions.

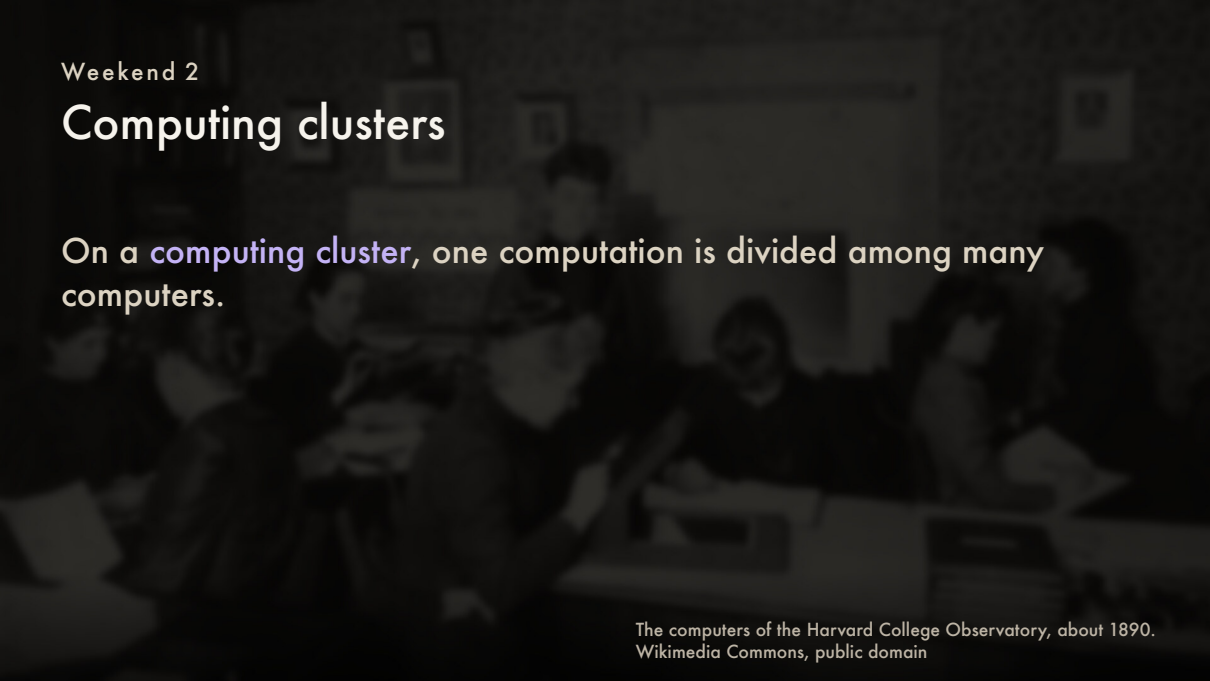


The computers of the Harvard College Observatory, about 1890.
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Weekend 2

Computing clusters

On a computing cluster, one computation is divided among many computers.



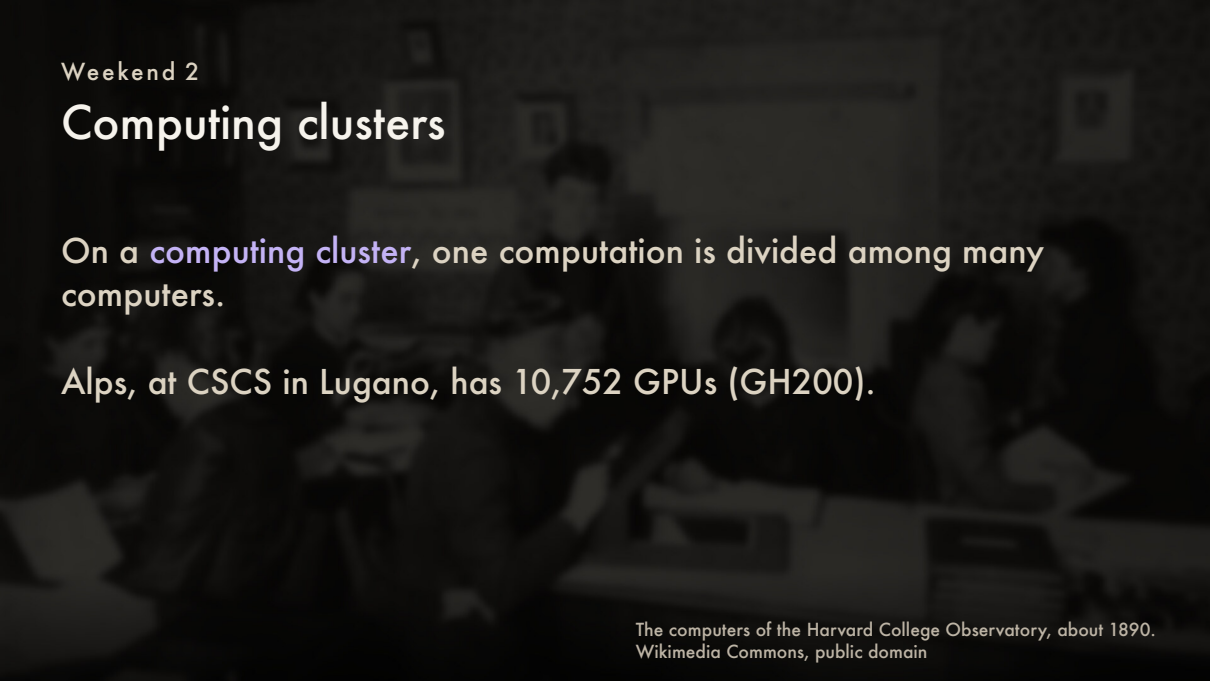
The computers of the Harvard College Observatory, about 1890.
Wikimedia Commons, public domain

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Alps, at CSCS in Lugano, has 10,752 GPUs (GH200).



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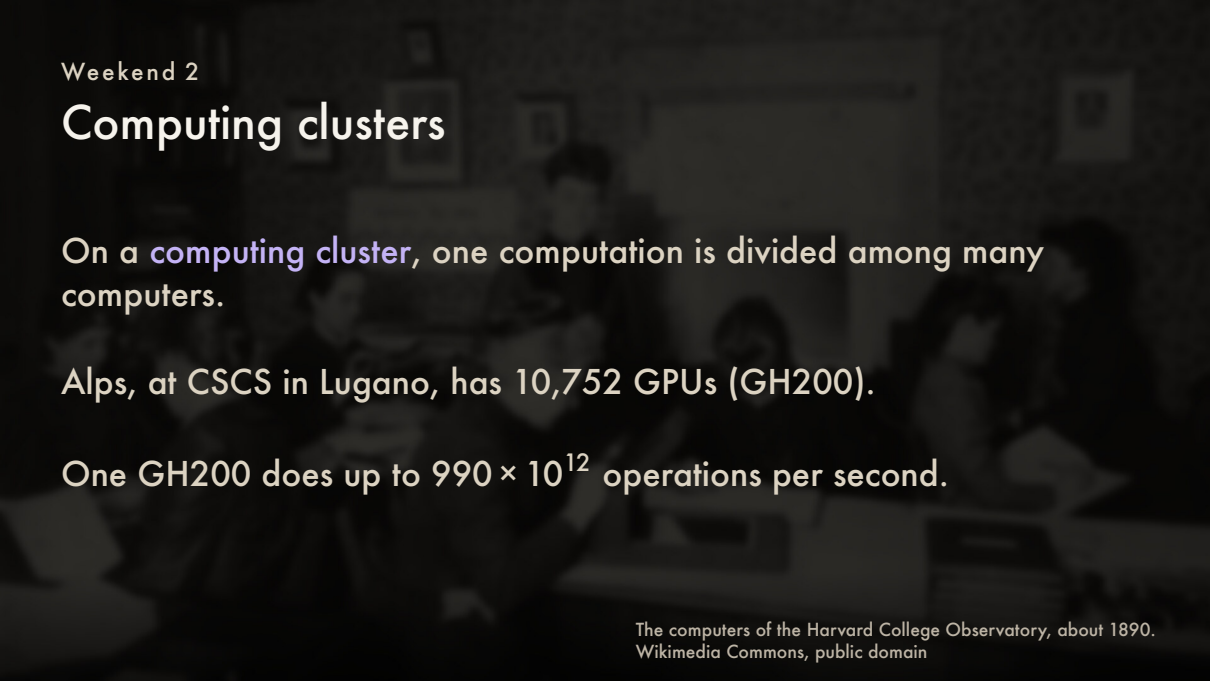
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One GH200 does up to 990×10^{12} operations per second.



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Weekend 2

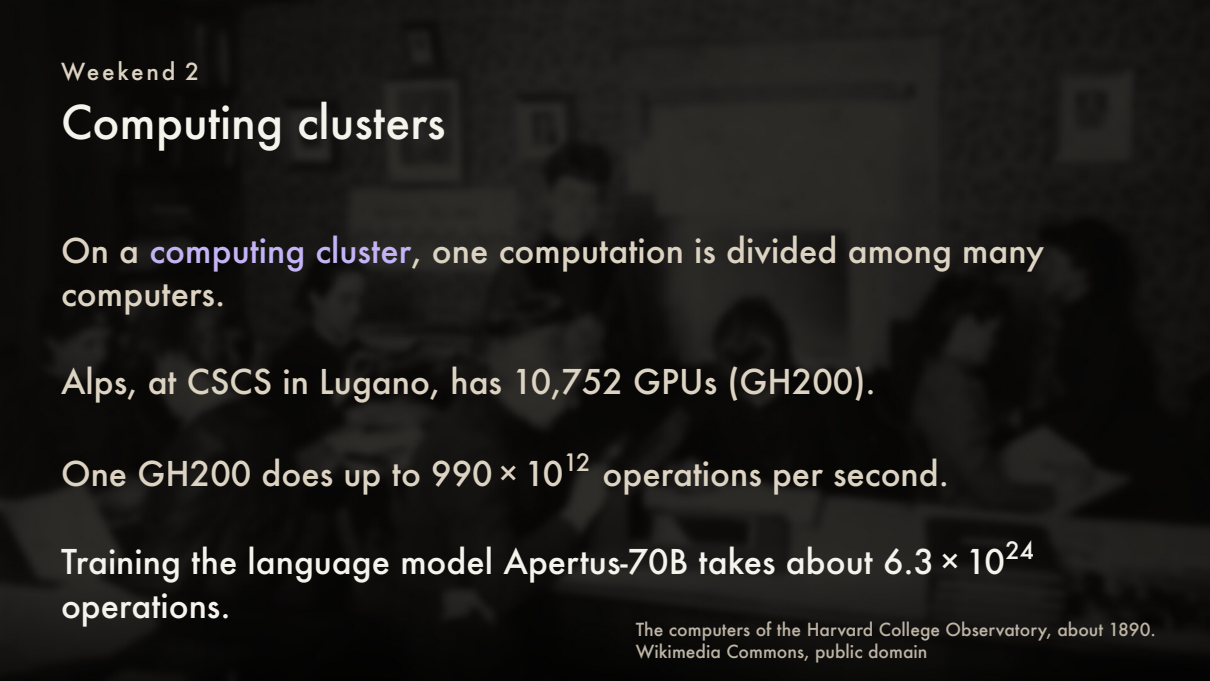
Computing clusters

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Training the language model Apertus-70B takes about 6.3×10^{24} operations.

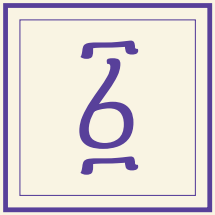


The computers of the Harvard College Observatory, about 1890.
Wikimedia Commons, public domain

Business case

Large-scale computing

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 13 November 2026 𓂀𓋹𓂀𓆑



Section 1

LinkedIn’s ads

LinkedIn's ads: the chance of a click

The chance that a member clicks an ad enters the ad auction, and the price depends on it.

Advertisers choose which clicks they pay for: some count likes or comments, others only visits to their website.

Borisyuk and others (LinkedIn), KDD 2024.

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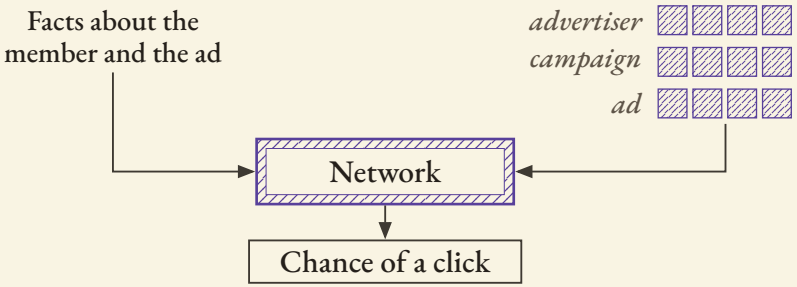
Over 1 billion

members

Borisyuk and others (LinkedIn), KDD 2024.

Numbers for every advertiser, campaign and ad

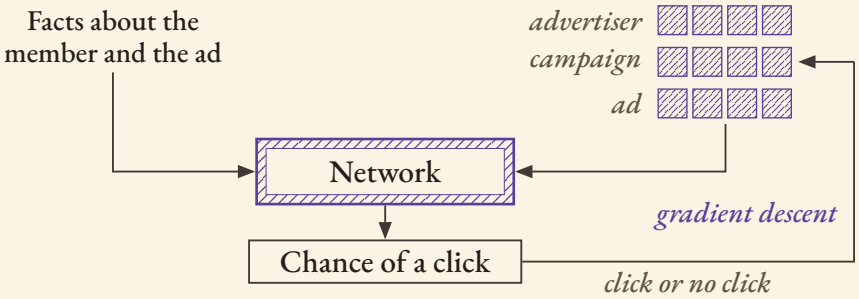
The matchmaker learned lists from ratings; here, from clicks.



Borisyuk and others (LinkedIn), KDD 2024.

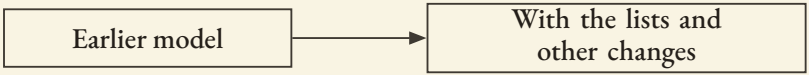
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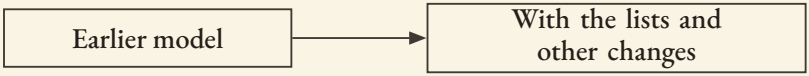
Borisyuk and others (LinkedIn), KDD 2024.

The value: A/B tests



Borisyuk and others (LinkedIn), KDD 2024.

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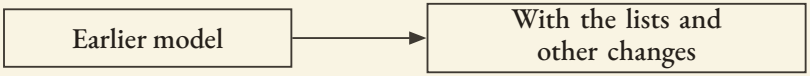
+4.3%

click rate on ads, relative change

LinkedIn put the new model into production.

Borisyuk and others (LinkedIn), KDD 2024.

The value: A/B tests



+4.3%

click rate on ads, relative change

LinkedIn put the new model into production.

−96%

training time in an offline test, when each new model builds on the earlier ones and learns from the newest half day of data

Borisyuk and others (LinkedIn), KDD 2024.